R for Reusability session

21 February 2025

Dear participants,

Welcome to the R for Reusability session, as part of the FAIRFest event organized by the FAIR-IMPACT European project and FAIRCORE4EOSC European project.

Thank you for your engagement and contributions to this FAIRFest session.

Kind regards,

R panel organizing committee

R for Reusability session
- License
- General Information
  - Links
  - Participation Guidelines
  - Guidance
- Agenda
- Ice-Breaker: Meet your neighbor online
- Activity 2: Do we care about Reusability?
- Activity 3: Reusability scope
- Activity 4: What you always wanted to ask and never dared to?
- Conclusion: Handing over: and now what?
- Feedback form
- R Panel organizing committee (FAIR-IMPACT/FC4E):

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General Information

Links

- Zoom: anonymized
- FAIR-IMPACT Website: http://fair-impact.eu/
- FAIRCORE4EOSC Website: https://faircore4eosc.eu/
- Event page: https://fair-impact.eu/events/fair-impact-events/fairfest-celebrating-advancements-fair-solutions-eosc
- Slides for the session: https://fair-impact.eu/sites/default/files/2025-02/0.%20%28ALL%20PPTs%29%20FAIRfest_Reusability_session%281%29.pdf

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Participation Guidelines

- In full:
    - **Code of Conduct**
        - We do not tolerate harassment in any form.
        - All communication should be appropriate for a professional audience.
        - Be kind to others. Do not insult or put down other participants.
        - Behave professionally in all communication channels (Slack, Zoom, Notes, etc.)
        - We want to promote open but respectful discussion around our themes. Please be mindful of the language and tone that you use, as participants will have differing points of view and lived experiences.
        - Attendee Procedure For Reporting Harassment
- Privacy Policy
    - Social Media Policy
    - Intellectual Property Policy
        - Documents produced during the workshop (including but not limited to session notes, blog posts, Collaborative Ideas and presentations) will be shared after the event under a CC-BY 4.0 license. Please be mindful of what personal information you are willing to share in these.

Guidance

(Taken from the CW, SSI workshop)

- 🌼 Be kind. Be understanding. Be inclusive.
- 🚫 Please **do not Tweet** or share publicly any links to the Zoom room
- 👍 Please **do Tweet** otherwise using the hashtag
- 📹 the workshop will be recorded and the videos will be made available on the FAIR-IMPACT YouTube channel after the event (activities in breakout rooms will not be recorded).
- 😊 Turn on your video if you don't mind sharing your face (or off if you do!).
- 😶 Keep yourself muted when not speaking to minimise background noise.
- 👩‍💻 General chat in the slack should be prioritized over the zoom chat.
- 📝 In this document, please:
    - Feel free to add notes during the workshop in their associated sections.
    - Note that many people may be using the collaborative note-taking documents at the same time, so please be patient and understanding if there is a lag or a lot of movement going on at once.
- 🗣 Do not aggressively speak over others or overly dominate the conversation in a group session, be inclusive of everyone.

Agenda

Time (CET)	Programme
9.00	Welcome and introduction
9:05	Neighbor introductions
9:20	Panel member introductions
9:25	Panel and audience discussion
10:20	Wrap-up

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Ice-Breaker: Meet your neighbor online

Share your name, affiliation, EOSC projects and answer the questions:

- **What object type(s) do you work with?**
- **What is your experience with FAIR?**

Name	Affiliation	In EOSC projects?	What object type(s) do you work with?	What is your experience with FAIR?
Margreet Bloemers	ZonMw	In EOSC funders forum	DMP, FAIR Implementation Profiles (FIP)	FIPs, developing a FAIR Funding Framework on basis of nanopublications; fellow at GoFAIR Foundation
Anastasiia Iarkaeva	BIH@Charité	No	Open and restricted data sets, Open code in biomedicine	FAIR Data assessment
Joyce Nijkamp	WUR	No, would like	Enterprise architecture	Architecting FAIR ecosystems
Radoslava Kacová	Masaryk Memorial Cancer Institute	no	Sensitive biobanking data	Developing metadata catalogue to offer data derived from biological sample to researchers
Estelle Nivault	Inria	FAIRCORE4EOSC - WP6, in particular task 6.4 on metadata and PIDs for software: Curation and standardization	all research work (publications and software research source code)	HAL software repository and interoperability with SWH
László Kovács	SZTAKI	FAIR IMPACT	RO-Crate, DMP, ELN, CRIS, Dataverse, ...	develop research infrastructures and mission critical services
Mike Priddy	DANS	FAIR-IMPACT/FAIRCORE4EOSC		It is still not widely known, or at least not widely enough. However it is being picked up

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Name	Affiliation	In EOSC projects?	What object type(s) do you work with?	What is your experience with FAIR?
				beyond research in areas such as governmental data
Joy Davidson	DCC	FAIR-IMPACT, Skills4EOSC, FIDELIS	People ;)	Helping to raise awareness among stakeholders on why FAIR matters. Developing the technical infrastructures and standards to enable FAIR are crucial but so too is engaging effectively with the people (researchers, data stewards, service providers, policymakers, governments, etc.) who will use them, value them and - importantly - be willing to pay to sustain them (directly or indirectly).
Marie Schmit	Institut Pasteur	No	Scientific workflows and annotation standards (RO-Crate, various ontologies...)	Working on making scientific workflows more FAIR by annotating them, using existing standards like ontologies, schemas or controlled vocabularies.
Elodie Chapeaublanc	Institut Pasteur	No	Develop scientific web applications and tools	Using FAIR principles at severals levels of development of tools : gitlab, github, readme files, documentation, license, HAL, Software Heritage, ...
Mari Kleemola	FSD	OSTrails, FIDELIS	Data, metadata, standards, policies, best practices	Using FAIR principles to improve practices and policies; assessing repositories; understanding the gaps
Maaike Verburg	DANS	FAIR-IMPACT, FIDELIS		Training data stewards on FAIR and its

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Name	Affiliation	In EOSC projects?	What object type(s) do you work with?	What is your experience with FAIR?
				implementation
Daniel Turner	DCC	FAIR-IMPACT, Skills4EOSC	Policies, training materials, raising awareness	Seeing it grow in usage in cultural heritage and archaeology alongside CARE
Maja Magel	NFDI4Microbiota	no	Training material, raising awareness, metadata standards for (micro)biological life sciences, ELN	Introducing actionable FAIR items to data collectors needs to happen gradually (metadata fields, schemas, ontology terms) before introducing highly technical infrastructure requirements
Martina Stockhause	IPCC-DDC	no	FAIR, TRUST, traceability/provenance in support of transparency/trust into organizations like IPCC	The principles described in FAIR were guiding us as a long-term archive before they were introduced. FAIR has pushed and guided the discussion but there are several perceptions/interpretations.
Peter Mutschke	GESIS	no	Research data, research software	Developing FAIR-compliant research data infrastructures

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Name	Affiliation	In EOSC projects?	What object type(s) do you work with?	What is your experience with FAIR?

Activity 2: Do we care about Reusability?

Do you care about Reusability?	Why?
	Reusability has an aspect related to return on investment since improved reusability of research outputs directly translates into wider and more efficient use of publicly funded assets. I haven't looked very widely, but this is something for serious study and analysis. Without policy support data on this aspect, we cannot determine how much effort should be invested in ensuring reusability. Some research outputs are a bit simpler to assess (code), some will be very difficult to
Margreet Bloemers: yes	By creating FAIR Implementation Profiles, a research community (FIC) decides and shows (in a machine actionable way) what standards it uses, enabling other researchers in the same research domain to use them as well > find them on Search FAIR Supporting Resources - FAIR Connect Pro > facilitating convergence/harmonisation > stimulating reuse of standards and tools. >> thereby supporting reusability of data and other resources.

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Do you care about Reusability?	**Why?**
Joyce Nijkamp: Yes	Sustainability in various dimensions
Anastasiia Iarkaeva: Yes!	Otherwise why do we publish all the data/software and what is the benefit of OS? In my field, training the researchers on the appropriate licensing, as well as what to pay attention to when choosing a data repository is facilitating a proper reuse. However, just putting the data online does not help to re-use it, but with well-prepared metadata others could potentially save time and/or resources by just building their research upon existing sources.
Mari Kleemola: Yes	If something is not made reusable or kept reusable, why use scarce resources to store or preserve it?
Yes	if only for environmental reasons. Each phase of the lifecycle thus introduces its own cost: storage, transport on networks, handling with computers, back-up and, finally, long-term archiving, etc. As soon as we have validated that the data produced or acquired is worth keeping (the first good environmental approach being to neither produce nor keep useless data)
Elodie Chapeaublanc : yes	don't reinvent the wheel but often data and informations are missing. Reusability is helpful for the sustainability of tools
P:yes	Saves time, effort and resources and why not build on it !!! unless there is less to build and you like to re-invent the wheel !!!
Joy Davidson: yes	Remember that the data you reuse may be your own :) Good RDM leading to FAIR data helps you reuse your own data over time as well as helping others to benefit from it.

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Do you care about Reusability?	**Why?**
Mike Priddy: yes, but	Not all researchers like their data to be reused - in certain disciplines it is how their careers are built.
Maaike Verburg	Is caring enough? How do we turn caring about something into doing something about it?
margreet	People who take care of the reusability should be recognized for their efforts. E.g. by coupling you orcid to the resource you created/share.
Daniel Turner	Field archaeology often operates through destructive methods like excavation - sites and features that are excavated to mitigate impact from modern infrastructure on the cultural heritage record are forever lost apart from what is recorded during and directly after field operations. If that data is not 'permanently' reusable, it is all for nothing.
Maaike Verburg	Reusability also has levels: is something reusable by your direct colleague who knows how you work already? Is it reusable to someone in your field who doesn't know you? Is it reusable for different purposes or disciplines? What information is needed for each level of reusability and how does this then relate to accessibility?
	Data producing organizations become data hoarders and they wont let any one touch it and then it gets political…. I will not do it.. Will not let you do it either.

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Do you care about Reusability?	**Why?**

Activity 3: Reusability scope

How do you see Reusability ?	**Are the FAIR principles enough for reusability? Is it crucial to do Open Science?**
(Data) Reusability is the elephant in the room, since it requires • additional effort to encode and publish data subject to more constraints (formats and schema, vocabularies, ...), • and/ or add more metadata. DANS has started formulating a readiness scheme that goes beyond FAIR, defined in summary in the table.	
	FAIR can only work if people facilitating it are well aware of it and believe in it. Many folks talk about it like a badge of honour and lack the knowledge or skills that actually make it happen. So that is the biggest barrier as there are more talkers than doers !!! Some management really sit in ivory towers as it's called in this world !!!
	I think that the discussion shout resuability should be coupled more often with deletion of data too. Researchers are often unwilling to better describe their data because writing the metadata is time consuming but also by not doing so they risk making their data unusable and with finite space it risks being deleted in the future.
I don't think it's easy to force researchers to make their data FAIR. The only way is to include	The support staff are making considerable efforts to raise awareness, train and inform, and we need to be pugnacious. It's utopian to think that this is enough. Only

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How do you see Reusability ?	Are the FAIR principles enough for reusability? Is it crucial to do Open Science?
the FAIR principles in the evaluation criteria.	a legal and political framework will change things, the rest is “only” technical.

Activity 4: What you always wanted to ask and never dared to?

- Q: **If identification tools (DOI or others) are not available, what methods do we have to track the reuse of data ?**
    - A: Personally, I have put research data in places like YouTube & Flickr where it has outlasted the national research repository. It can be tracked, but this is not normally a place to share.
    -
    -
- Q:
    - A:

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- Q:
  - A:
- Q:
  - A:

Conclusion: Handing over: and now what?

Herve L'Hours: Reusability is an incredible target to hit, you need to define what you are working towards, how, and when you'll know you've reached it.

Elena Breitmoser: There is a need for FAIR and beyond for your object. But how do we make sure we don't diverge too much in the disciplines once FAIRCORE4EOSC and FAIR-IMPACT end? How do we build a community for sharing input and preventing reinvented wheels?

Daniel Garijo: There is a clear importance of documentation into the future. Usage examples are especially essential, even though they're out of the direct scope of FAIR. For better reusability into the future, ask if any reuse has been successful before publication, before you reuse an object.

Alex Ioannidis: As a service provider, think about user experience. How do we work in the background to capture the needed information without burdening researchers? But also, educating researchers is still important at the same time.

Wim Hugo: Certified repositories are only the tip of the iceberg. There are about 150 certified repositories worldwide currently, but there are about 3000 existing repositories captured in re3data and even more repositories that likely exist outside of that. The challenge ahead is quite enormous to get these good practices across a much larger order of magnitude.

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R Panel organizing committee (**FAIR-IMPACT**/**FC4E**):

- Maaike Verburg (DANS)
- Morane Gruenpeter (Inria/Software Heritage)
- Mike Priddy (DANS)
- Neil Chue Hong (UEDIN-SSI)
- Elena Breitmoser (EPCC, University of Edinburgh)

R panelists

- Wim Hugo (DANS-KNAW) => CAT
- Hervé L'Hours (UK Data Service) => Repository Transparency/Prototype
- Alex Ioannidis (CERN) => RSAC Zenodo/InvenioRDM
- Elena Breitmoser (EPCC) => F-UJI for Research Software
- Daniel Garijo (UPM) => FAIR Semantic artefact assessment (O'FAIRe + FOOPS)

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